

ER DIAGRAM

Entity Weak Entity Participation Constraints Attributes Relationships

Entity Strong Entity
Simple 1 : 1 Composite 1 : M

Depends on Strong
Total

Partial

M : N

Recursive

Multivalued

Derived

DATABASE SYSTEM

SCHEMA INSTANCE

Changes
frequently

Conceptual schema logical schema physical schema

LOGICAL DATA PHTSICAL DATA

Changes to the
conceptual (logical)
schema do not
affect user views or
application
programs.

Changes to the
physical storage do
not affect the logical
schema or
application
programs.